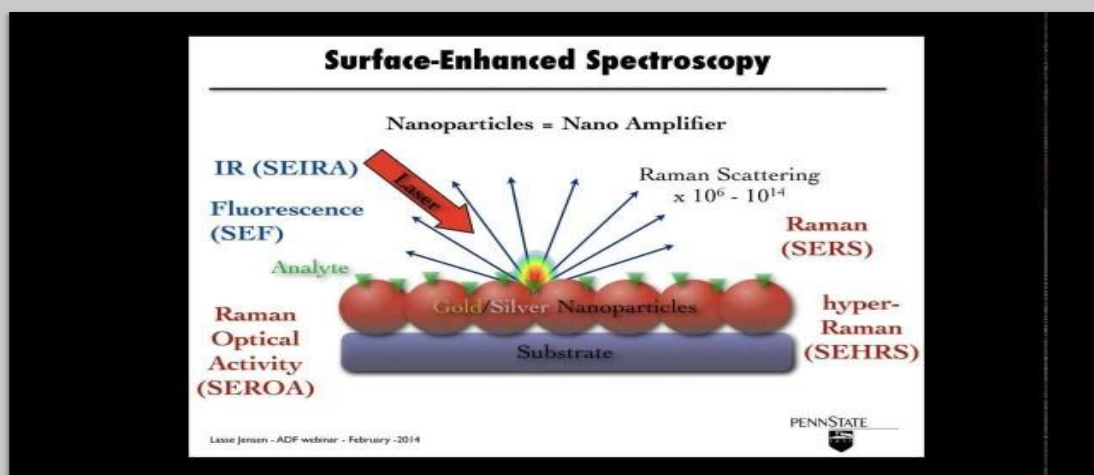

*FUNDAMENTALS OF RAMAN SPECTROSCOPY AND
SURFACE ENHANCED RAMAN SPECTROSCOPY (SERS) AS
AN ADVANCED POTENTIAL TOOL FOR DIAGNOSING
INFECTIOUS DISEASES AND OTHER DISORDERS*



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REVIEW

FUNDAMENTALS OF RAMAN SPECTROSCOPY AND SURFACE ENHANCED RAMAN SPECTROSCOPY (SERS) AS AN ADVANCED POTENTIAL TOOL FOR DIAGNOSING INFECTIOUS DISEASES AND OTHER DISORDERS

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ABSTRACT

The dispersion of light into component colours is termed as Spectroscopy. The ultimate objective of using spectroscopy is to realize the interaction of light with matter and the ways that can be used to quantitatively recognize the sample. Raman Spectroscopy is a spectroscopic approach which is utilized to examine rotational, vibrational, and other low-frequency modes in a system. This phenomenon depends on Raman scattering (which is also known as inelastic scattering of monochromatic light). Raman spectroscopy has multiple additional benefits for biomedical uses, including being relatively fast to acquire, and provides information at the molecular level and non-destructive. The concept behind Raman spectroscopy is that the radiation may get reflected, absorbed, or scattered hence the monochromatic radiation should be passed through the sample in such a manner so that the above inference can be obtained. Mode of action of Raman effect is described by two theories (1) Classical Theory of Raman Effect (2) Quantum Theory of Raman Effect. The Stokes shift term can also be used in Raman spectroscopy where it explains whether the Raman dispersed radiation is at higher energy (anti-Stokes shifted) or lower energy (Stokes shifted) than the Rayleigh dispersed radiation. In anti-Stokes Raman scattering the contrary happens, with the molecule ejecting a quantum of vibrational energy during the dispersing process and the anti-Stokes radiation, henceforth it has a shorter wavelength than the incident. In recent times the bioanalytical surface-enhanced Raman spectroscopy (SERS) has bloomed into a rapid-developing research field. SERS is a productive scientific method that has excellent prospective in diagnosis and bioanalysis, as denoted by its expanding utilization *in vivo*. SERS recognized strategies fall into two major classifications, indirect and direct. SERS can be used to diagnose several disorders falling under categories of bacterial contamination, cardiovascular & neurological oriented problems, diabetes, viral contamination, and different forms of cancer like breast, lung, prostate, colorectal oral and few additional cancers as well. SERS has demonstrated to be an excellent candidate for observing neurotransmitters and additional biological molecules connected with neurological disease with the utilization of metal nanoparticles. Label-free SERS sensors have been engaged to observe the preparation of dopamine with the detection of dopamine, phenylalanine, tyrosine, and norepinephrine. Integrating SERS with multivariate analysis permits for subtle dissimilarities in spectra due to variation between different molecules or disease and normal tissue states to be explained, which builds up the ability of SERS for disease diagnosis. In recent times, much progress has been made regarding the evolution of a biosensor that utilizes Raman spectroscopy to observe diabetes in real time. The most used glucose scanner need blood to be pinched from the finger and laid down on a strip, where a reaction occurs between the glucose oxidase (GOx) and blood glucose, and an electrode estimates the potency of the electric field produced. From the direct diagnosis of glucose by boronic acid arrest label molecules, ligands, and sandwiching to the indirect sensing of glucose through its catalyzed oxidation, there are many chances to implement SERS in the clinical management of glucose levels. These methods are likely to progress the scope, longevity, and convenience of continuous glucose checking technology and enhance the lives of diabetic patients. SERS is very constructive for the quantification and identification of biomarkers of cardiovascular diseases, such as uric acid. C-reactive protein (CRP) is another indicator that has recently been observed as a potential marker of cardiovascular disease, using an Au nanoparticle-based SERS bioassay for diagnosis. In this bioassay, AuNPs are operationalized with an antibody for CRP, along with horseradish peroxidase (HRP) and leucomalachite green (LMG). *In vivo* SERS diagnosis techniques have been evolved to detect and assisted with drug delivery in patients with cardiovascular diseases. Atherosclerosis is a cardiovascular disease which creates plaque to develop in arteries. SERS has been utilized for both treatment and early detection of this condition. SERS has been applied to detect viruses including Hepatitis, influenza, HIV and those answerable for tropical diseases.

A scientist has made a SERS-based lateral flow immunoassay (LFIA), applying influenza A nucleoprotein as a selected molecule. Hepatitis is a major source of liver diseases, which includes liver cancer, ensuing the need for a diagnosis technique. A scientist evolved a SERS-based immunoassay with multifold spotting of the two viral diseases by using SERS-encrypted Au nanostars integrated to antibodies specific to each virus, and with separate Raman reporter molecules in an immunoassay to differentiate between nonstructural protein 1 (NS1) biomarkers. The technique was tested against a negative control of mosquito-borne chikungunya virus, with no Raman signal acquired from the negative control. SERS is an enchanting target for accomplishment in cancer diagnosis due to high sensitivity, little-to-no sample clarifying, and rapid detection times. The most lethal cancers in men include liver, colorectal, lung, prostate and stomach cancers; for women, the most lethal cancers are lung, colorectal, cervical, stomach and breast cancers. A SERS-based technique termed Raman-encrypted molecular imaging (REMI) has been evolved to grant for the sensitive, rapid and specific resolution of cancer cell-positive surgical margins during tumour removal policies. SERS and PCR methods were united for the tandem multiplexed diagnosis of single-base abnormalities located in melanoma lesions to distinguish between three varieties of point mutations in tumor DNA. SERS grants for the constant, extremely sensitive diagnosis and calibration of different biomarkers and end products of ailments states, which marks it as an excellent choice in the identification and treatment of many disorders that are global public health apprehension. The capability to establish substrates from gold, which is non-virulent during in vivo application, is a supplementary welfare supplied by SERS. SERS is likely to assist the scope, longevity and convenience of constant glucose checking technology, as well as assistance in earlier diagnosis of neurological, viral, cardiovascular diseases, and cancer. It is believed that SERS will resume to remain at the forefront of highly sensitive analysis in both in vivo and in vitro utilization.

KEYWORDS: Spectroscopy, Dispersion, Spectrum, Technique, Analysis, Diagnosis, Radiation, Wavelength, Biomarkers, Lethal, Virulent, In vivo, In vitro, Sensitivity, Specificity, Drug delivery, PCR, Nanoparticles, Mutations, Neurological, Glucose, Cardiovascular, Cancer, Viral, Global, Public health

INTRODUCTION

The dispersion of light into component colours is termed as Spectroscopy. It can also be defined as a technique for measuring the amount of light absorbed and the intensity of light passed through a chemical substance. According to analytical science, a unique characteristic spectrum is present in every element or compound. Light is dispersed by each compound over a specific range of wavelengths. [2] The ultimate objective of using spectroscopy is to

- ✓ Realize the interaction of light with matter and the ways that can be used to quantitatively recognize the sample.
- ✓ Fulfil different purpose like monitoring diffused oxygen content in aquatic ecosystem and freshwater, discovering the atomic structure of a sample, discovering the metabolic structure of a muscle, the spectral emission lines of distant galaxies can be studied, changing the design of drugs to enhance the potency, characterization of proteins, respiratory gas analysis in hospitals, space exploration. [3]

Individual variety of spectroscopy exists like 1) Raman spectroscopy, 2) Photoemission, 3) Time-resolved, 4) Circular Dichroism, 5) X-ray photoelectron, 6) IR Spectroscopy (Infrared spectroscopy) 7) Acoustic resonance.

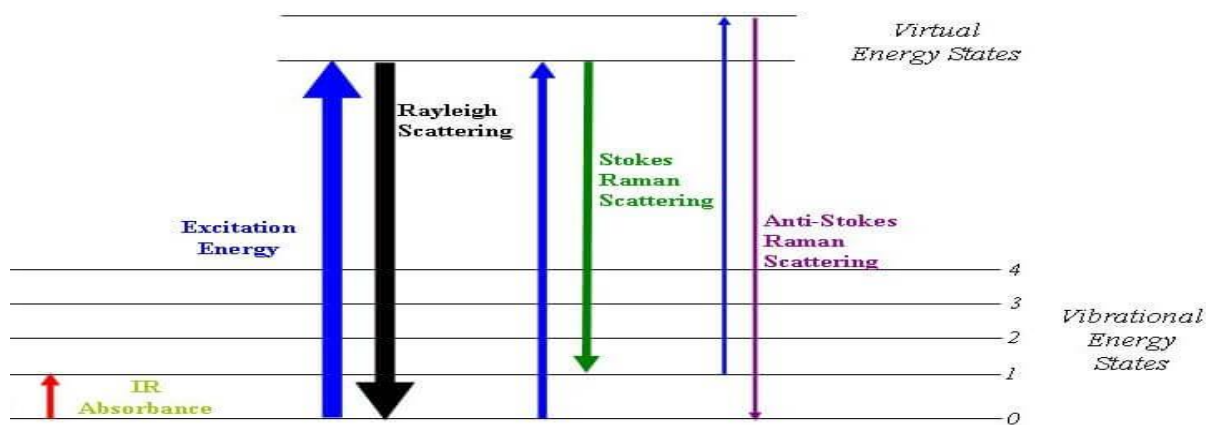


Fig 1: Diagram showing phenomenon of Raman spectroscopy from Ref. [2]

Raman Spectroscopy is a spectroscopic approach which is utilized to examine rotational, vibrational, and other low-frequency modes in a system. Raman's spectroscopy is frequently utilized in the field of chemistry to deliver a fingerprint which in turn can be used to recognize molecules. [1] The name of this phenomenon has derived from the name of famous scientist Sir C. V. Raman. This phenomenon depends on Raman scattering (which is also known as inelastic scattering of monochromatic light). Shifting of energy of the laser photons occurs up & down as the light interacts with the molecules or phonons of an object. [1,2] Vibrational modes of an object or system are formed due to this up down shift of laser photon. Other modern kinds of Raman spectroscopy are resonance Raman, surface-enhanced Raman, polarized Raman, tip-enhanced Raman, transmission Raman, stimulated Raman (analogous to stimulated emission), hyper Raman and spatially offset Raman. [3]

The incoming photons are absorbed or scattered when a sample is illuminated with a light. If absorbed, the photon energy is transmitted to the molecules, whereas if scattering of a photon takes place and the energy is maintained, it is termed as elastic scattering. However, a small segment of dispersed photons (1 in every 10 billion photons) can be scattered inelastically, this minute energy difference between the scattered and the incident photon is the Raman effect. [4] Raman spectroscopy has multiple additional benefits for biomedical uses, including being relatively fast to acquire, and provides information at the molecular level and non-destructive. In addition to this water gives rise to faint Raman scattering, which means the existence of water in the sample does not cause an obstacle for the spectrum that is being tested. [4] The main drawbacks of Raman spectroscopy comprise the very faint Raman signal and the existence of unpleasant noise sources such as the severe fluorescence framework that exists in biological samples. [2]

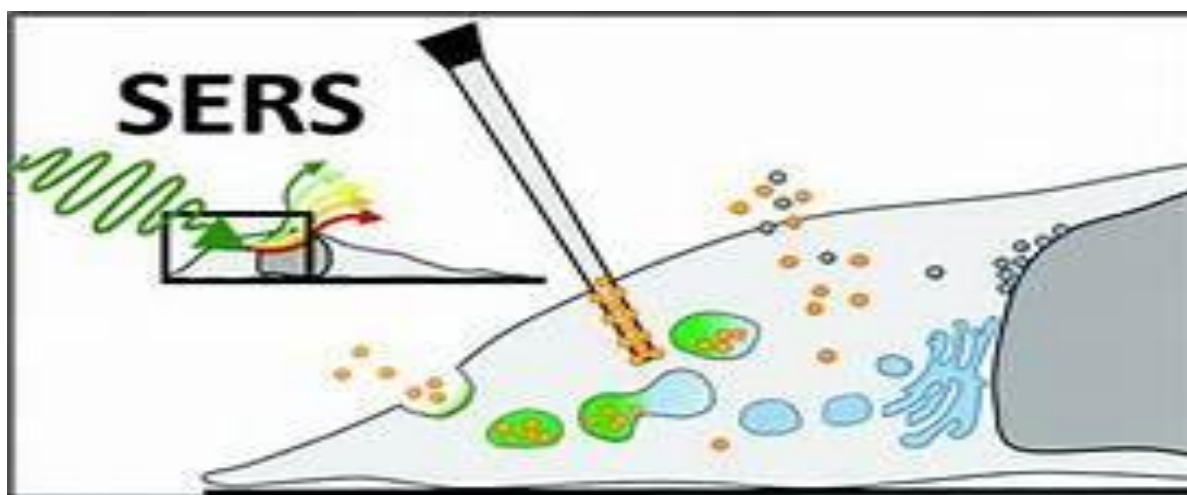


Fig 2: SERS-an advanced potential method for diagnosis of several disorders from Ref. [11]

SERS (Surface Enhanced Raman Spectroscopy) can achieve single-molecule (SM) detection henceforth it is regarded as an ultrasensitive technique. The enhancement (by factors up to 10^8 or even larger) of inelastic light dispersing by those particles that are adsorbed on the top of the nanostructured metals, such as silver or gold nanoparticles causes the sensitivity of SERS. [14] SERS also provides structural details in biological media in surrounding environments. SERS is user-friendly, in contrast with other highly sensitive methods (like for example mass spectrometry) and is relevant to aqueous solutions without the requirement for complicated sample preparation protocols. SERS is rapidly becoming a scientific method of option for successful utilization in disease detection and diagnosis and bioanalysis. [9] Additionally, SERS permits disease detection inside living organisms (*in vivo*) after a layer of biocompatible SERS-operative nanoparticles or substrates with suitable protective shells or surfactant. There are two principle perspective when using SERS for bioanalyte detection, namely label-free SERS detection and label-based detection with SERS tags. [11]

FUNDAMENTAL PRINCIPLES OF RAMAN SPECTROSCOPY

The concept behind Raman spectroscopy is that the radiation may get reflected, absorbed, or scattered hence the monochromatic radiation should be passed through the sample in such a manner so that the above inference can be obtained. The frequency of scattered photons is different from the incident photon as the rotational and vibration property vary. This causes a change in the wavelength, which is examined in the IR spectra. [5]

The variance between the scattered photon and the incident photon is termed as Raman shift. When the energy related to the dispersed photons is lower than the energy of an incident photon, the scattering is termed as Stokes scattering. When the energy of the dispersed photons is higher than the incident photon, the scattering is termed as anti-Stokes scattering. [1]

MECHANISM OF RAMAN EFFECT:

Mode of action of Raman effect is described by two theories

(1) Classical Theory of Raman Effect

(2) Quantum Theory of Raman Effect

CLASSICAL THEORY: Polarizability of molecules is the basis of the classical theory of Raman effect, which review how easily an electron cloud of a molecule can be deformed by an electric field (light). The method is based on molecular distortion in electric field E detected by molecular polarizability α . The laser beam can be observed as an oscillating electromagnetic wave with electrical vector E . When it interacts with the sample it produces electric dipole moment $P = \alpha E$ which distorts the molecules. Because of recurring deformation, molecules begin vibrating with characteristic frequency. The dispersed light can have a prevalence equivalent to the incident light (Rayleigh), equivalent to the incident light plus the vibrational frequency (anti-Stokes) and equivalent to the incident light minus the vibrational frequency (Stokes). [6,8]

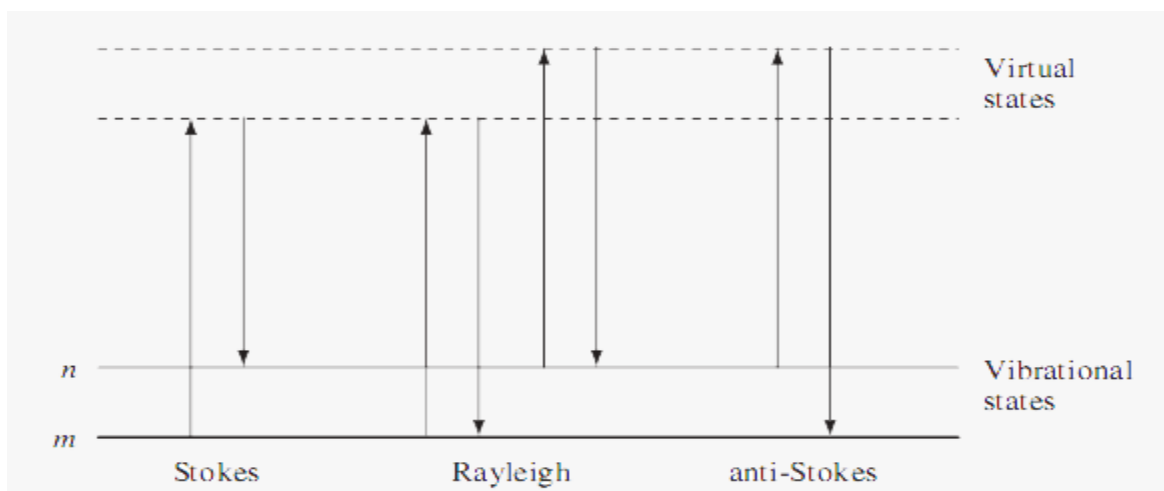


Fig 3: Diagram showing stokes, anti-stokes and Rayleigh phenomenon of Raman Spectroscopy from Ref. [8]

QUANTUM THEORY: Upon interaction of light with matter, the photons which comprise the light may be consumed or dispersed or may not interconnect with the material and may pass directly through it (Smith and Dent, 2005). When there is interaction between monochromatic light and the molecules, majority of the photons are dispersed without any alteration in energy. This process is termed as an elastic or Rayleigh scattering, and it happens when the electrons in a molecule swing (oscillates) in resonance with the administered electric field of the incident light. However, minute number of photons (1 out of 10^6 to 10^9) are inelastically dispersed and goes through a alteration in energy. The correlation between frequency, wave number, wavelength and the energy are as follows:

- $E = h\nu$ (ν is frequency in sec^{-1} and h is Planck's constant)
- $E = hc\nu$ (ν is wave number in cm^{-1} and c is the speed of light in cm/sec)
- $N = 1/\lambda$ (wavelength is represented in nm by λ) [8,6]

The inelastic dispersing process is termed as Raman effect (Raman scattering). The alter in photon energy happens because a molecule may tremble during that time and an electric field is applied which causes the electrons to oscillate in resonance. A vibrational mode that alters

dipole moment induced by electric field (molecular polarizability) causes a change in incident photon energy. The variance in energy between the inelastically scattered photons and the incident photons is called Raman shift. The representation of the intensity of the inelastically dispersed light as an outcome of the energy change is called Raman spectrum. [7]

STROKE'S SHIFT IN RAMAN SPECTROSCOPY

The Stokes shift term can also be used in Raman spectroscopy where it explains whether the Raman dispersed radiation is at higher energy (anti-Stokes shifted) or lower energy (Stokes shifted) than the Rayleigh dispersed radiation. [5,1] The majority of photons scatter molecule elastically with no change in the vibrational energy of molecule during the scattering process (Rayleigh scattering) when radiation is scattered from a molecule. In Stokes Raman scattering the molecule attains a quantum of vibrational energy from the photon during the dispersing process and the Stokes radiation, henceforth it has a longer wavelength than the incident radiation. [7] In anti-Stokes Raman scattering the contrary happens, with the molecule ejecting a quantum of vibrational energy during the dispersing process and the anti-Stokes radiation, henceforth it has a shorter wavelength than the incident. Raman summits are featured by their wavenumber shift elsewhere from the incident radiation, with Stokes summit having a positive wavenumber shift and anti-Stokes shifts being negative. [4]

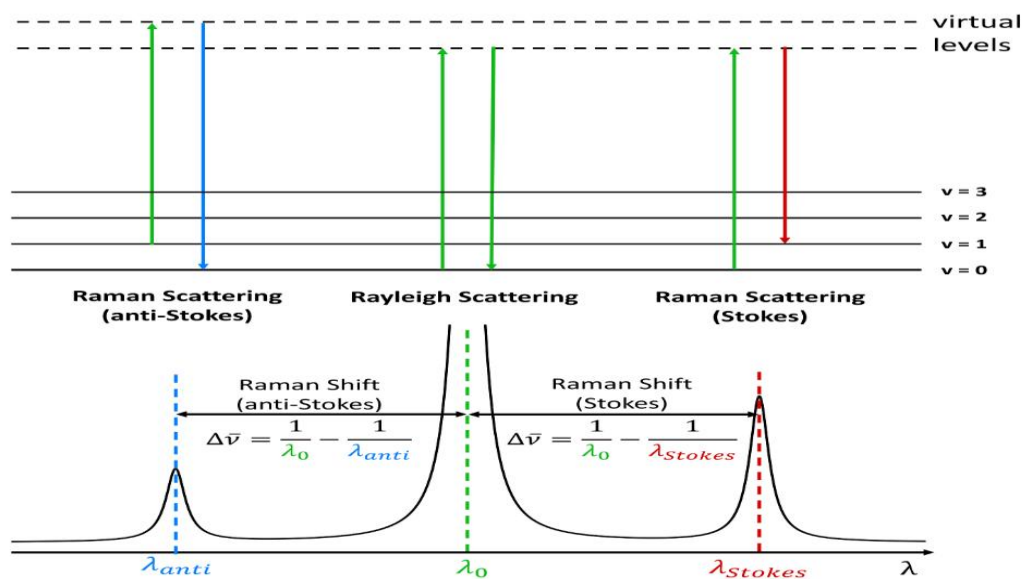


Fig 4: This diagram shows the demonstration of Raman Shift from Ref. [4]

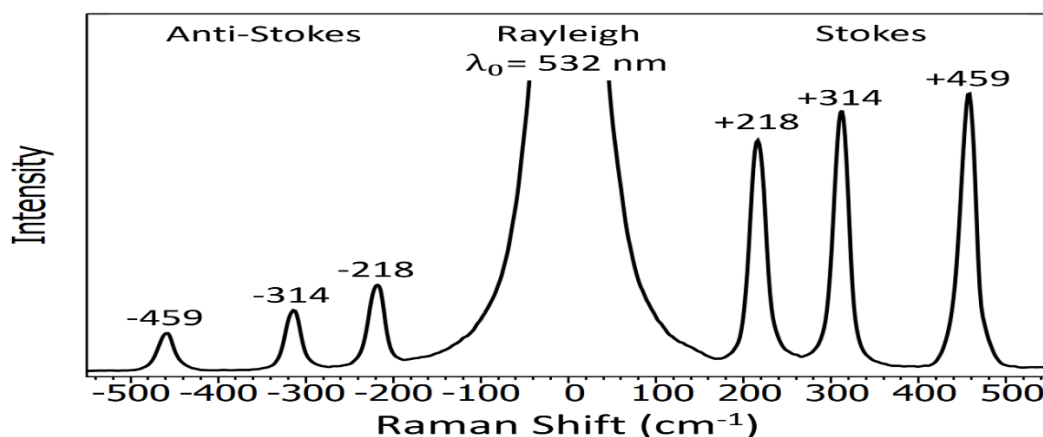


Fig 5: A model showing distinct stokes and anti-stokes shift in case of CCl₄ from Ref. [5]

A model Raman spectrum of CCl₄ is shown overhead with the Stokes and anti-Stokes shifts made distinct. Since anti-Stokes Raman scattering implies the deprivation of vibrational energy by the molecule and should originally be in one of the higher-lying vibrational levels ($v > 0$) for it to happen. [6] At equilibrium the maximum number of the molecules are in the $v = 0$ level and for that reason the anti-Stokes lines are remarkably fainter than the Stokes, as can be seen overhead, and for this reason, only the more intense Stokes lines are typically measured in a standard Raman experiment. [5, 8]

RAMAN SPECTRUM: To study the Raman effect, the wavelength of the dispersed photon is transformed to wavenumber. These wavenumbers are represented by the x-y plane. The wavenumbers are plotted along the x-axis and the Raman intensity is plotted on the y-axis. The variance between the intensity and the wavenumbers is known as the Raman spectrum. [1]

SERS (SURFACE ENHANCED RAMAN SPECTROSCOPY) AS A POTENTIAL ADVANCED TECHNIQUE FOR DIAGNOSIS

In recent times the bioanalytical surface-enhanced Raman spectroscopy (SERS) has bloomed into a rapid-developing research field. Due to its outstanding multiplexing ability and high sensitivity, SERS is a productive scientific method that has excellent prospective in diagnosis and bioanalysis, as denoted by its expanding utilization *in vivo*. [11,12] The major focus is on SERS-based detection methods of detection of pathogens as point-of-care solutions for early incurable disorder diagnosis and bacterial disease detection. In addition to it numerous favourable *in vivo* utilizations of SERS are observed. SERS recognized strategies fall into two major classifications, indirect and direct. Direct sensing depends on the binding or adsorption of an analyte molecule straight to a plasmonic substrate, and the evolving spectral characteristics correspond to the vibrational modes of the analyte itself. In the indirect sensing system, the SERS signal approaches from a reporter molecule (typically a strongly Raman scattering molecule or a dye [14]) rather than the analyte itself. When direct sensing is used, it can be tough to tally spectral intensity to congregation because SERS intensification is non-linear at elevated analyte concentrations; there is not always a constant adsorption of molecules on the surface of the nanoparticle; and wide spacing between particles causes

inefficient “hot spot” formation, results in reduced enhancement. In addition to it direct sensing in biofluids can develop spectra which are hard to explain due to enhancement of the constituents of the fluid itself [12,9,10]. Indirect sensing techniques can control these disadvantages by using molecular recognition agents, capture ligands or antibodies to selectively encapsulate the analyte molecules and initiate them adjacent to the enhancing surface. Indirect sensing does not investigate the molecule of interest itself, so we obtain no knowledge about the vibrational modes of the molecule. [13]

BACTERIAL DISEASE DETECTION

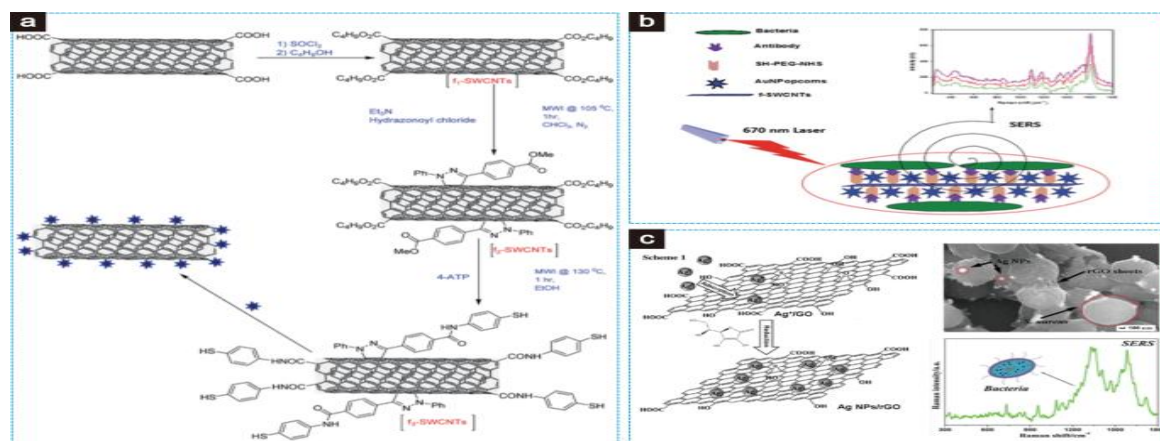


Fig 6: This diagram shows the nanoparticle-based SERS diagnosis of bacterial infections from Ref. [22]

Another fascinating example of the prospects of SERS is the spotting of bacteria. A scientist developed a particular policy to detect *Escherichia coli* and test antibiotic vulnerability. They were able to detect bacteria at a level of 10^2 CFU/ml with the use of rhodamine 6G-modified gold core–silver shell nanorods and confirmed their presence in spiked mouse blood and to distinguish antibiotic resistance in less than just 3.5 h. [22] Other than ultrasensitive spotting of bacteria, in SERS-based outlook, it is possible to go forward and to photothermally destroy bacteria. In a study a scientist coupled bacteria to the graphene oxide-modified tags, SERS diagnosed both Gram-negative (*E. coli*) and Gram-positive (*S. aureus*) bacteria and, photothermally obliterate them, showing the capability of *in situ* monitoring of the photothermal antibacterial reaction using a SERS trial. [22,23] SERS can be used to recognize particular frequencies of chemical outcome in complicated aquatic biological samples via appropriate clear and NIR laser probes. Based on metal colloids, self-assembled membranes, SERS biosensor technology and CNT templates has been advanced to upgrade the susceptibility and perfection. However, most of the perspectives still face the opposition of low replicability and low solidity. SWCNTs have the features of tangential Raman scattering and radial breathing mode. As a pillar for AuNPs, they may become a practical blended SERS substrate. [23] These constructions produce intense and severe peaks that are easily prominent from the fluorescent environment. To upgrade the stability and photothermal effects, f3-SWCNTs were selected as a template to produce Au nanopopcorn.

Selectivity is combined by targeting specific identification of monoclonal antibodies. A novel blended nanomaterial based on gold nanopopcorn-linked SWCNTs (AuNP@f3-SWCNTs) was manufactured successfully to obtain photothermal killing of bacteria by Hamme II's group and label-free detection. The application prospective of SWCNT/AuNP nanohybrid materials is amplified by the good combination with biological molecules and large aspect ratio. [26] A current challenge is to diagnose and accurately abolish antibiotic-resistant bacterial strains. Precisely, the outer membrane of Gram-negative bacteria supplies additional protective layers, such as against fungicides and antibiotics, which help bacteria combat antibiotics. To disclose this challenge, a scientist had constructed an Ag/ZnO/rGO nanocomposite as a versatile biomedical platform for SERS diagnosis and numerous killings of *E. coli*. [26,23] Superiorly, this work was the first review of integrating three materials for concurrent diagnosis and killing of bacteria. Recently, an environmentally friendly and simple synthetic technique for producing an effective SERS platform has attracted much awareness: A scientist suggested a fast and green method to produce AgNP/rGO composites and used the fabricated SERS substrate to diagnose bacteria such as *S. aureus* and *E. coli* with a perception sensitivity as low as 1×10^5 CFU/cm³. The environmentally friendly and simple in situ synthesis technique has potential implementation prospects in the construction of other graphene/metal matrix nanocomposites. [24]

NEUROLOGICAL DISORDERS DETECTION

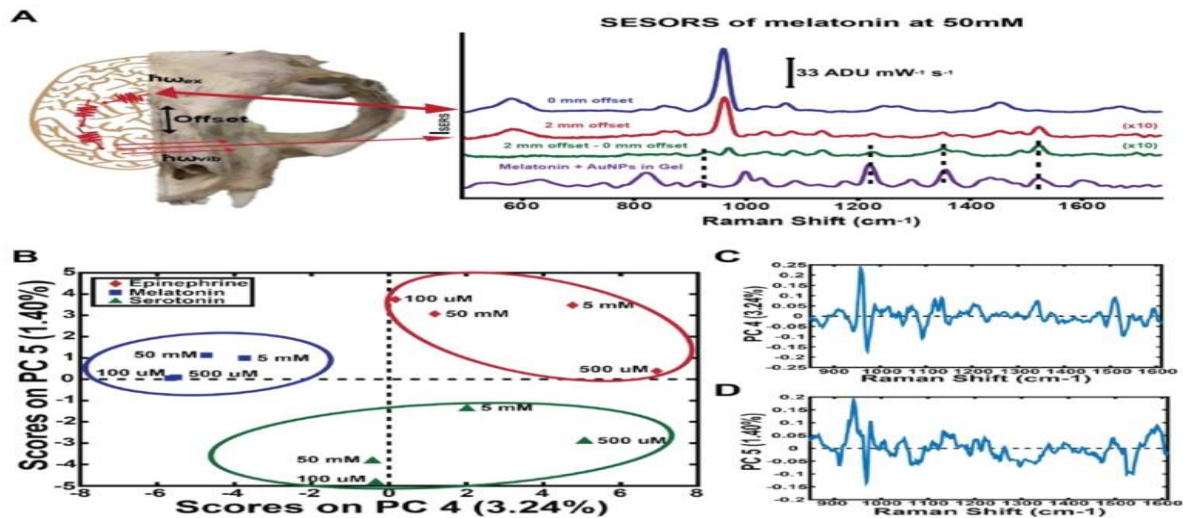


Fig 7: Light scattered by Raman theorem was secured at a point dimensional offset from the incident radiance point, recognizing the signal from the neurotransmitter to be obtained through a cat skull (A). PCA was utilized to demonstrate that the individual neurotransmitters are distinctly segregated from each other in a scheme of the scores of PC4 versus PC5 (B) and their matching loadings from Ref. [28]

SERS has demonstrated to be an excellent candidate for observing neurotransmitters and additional biological molecules connected with neurological disease with the utilization of metal nanoparticles. To regulate the feasibility of the utilization of metal nanoparticles in

biological organisation, toxicity studies have been guided for AgNPs and AuNPs. A scientist has found that the distribution of AgNPs in rats activated brain oxidative stress and alterations in amino acids and neurotransmitters. [17] Demyelination and astrogliosis of neurons related to neuronal vacuolation and degeneration were explored in histological examinations. They also found, however, that the simultaneous distribution of rutin compensated the toxic effects induced by the AgNPs. The consumption of AuNPs and their utilization as satisfactory nanoparticle sensors for subcellular SERS imaging was also examined, and it was found that the cellular consumption of the AuNPs relies upon the incubation time and size of the particles. The movement of AuNPs after consumption and clarifying through the endocytotic pathways was examined with particle spotting, characteristic transport velocities and intracellular diffusion coefficients. The results proved that AuNPs are acceptable intracellular probes for utilization in SERS, which takes advantage of the AuNP cluster in the cells to review their local chemical surroundings, making them a magnificent candidate for in vivo SERS diagnosis. The feasibility of SERS for the diagnosis of amino acid neurotransmitters such as glutamate and γ -amino butyric acid (GABA) was denoted by AgNPs, where limits of detection (LODs) of 10^{-4} M for GABA and 10^{-7} M for glutamate in aqueous solution were attained. These amino acid neurotransmitters are significant for neuroendocrine control and are related to epilepsy [27,28]. The detection of catecholamine and choline neurotransmitters as well as dopamine, acetylcholine and epinephrine with Ag has also been shown. A SERS substrate was prepared by AgNP electrodeposition using cyclic voltammetry onto tin-doped indium oxide (ITO) and size and surface coverage was managed by changing the rate and number of scans. The detection of dopamine is an evasive, yet highly advantageous, target due to the connection between the Parkinson's disease and loss of dopamine. SERS measurements have been gained on rat pheochromocytoma PC12 cells with Au nanopatterned substrates to ascertain the outcome of bisphenol-A, cyclophosphamide, and cisplatin on the level of extracellular dopamine liberated [17]. Very low LODs for dopamine have been attained using an AuNR dimer protected with an Ag shell (0.006 pM) and Au core-Ag shell nanoparticle-Au nanorod heterodimers (0.02 nM) [17]. Nanomolar LODs of dopamine and serotonin in restored body fluid have been attained using a graphene-Au nanopyramid heterostructure medium [27]. Label-free SERS sensors have been engaged to observe the preparation of dopamine with the detection of dopamine, phenylalanine, tyrosine, and norepinephrine [28]. The selective seizure of dopamine can be attained using metal nanoparticles modified with a dopamine-selective probe. This capture has been demonstrated with AgNPs altered with iron-nitrilotriacetic acid [27] and through a sandwich assay comprising of Au nanoplates altered with dopamine antibodies as one layer integrated with AuNPs encrypted with DNA and antibodies as the other layer [17]. The dosage of norepinephrine in dopaminergic cells can be diagnosed using magnetic dynabeads holding nanoparticles which are packed with DNA barcodes and antibodies that can sandwich the selective molecule [17]. AuNPs operationalized with Raman reporter molecules and antibodies have also been utilized for SERS diagnosis of the IgG protein. SESORS was utilized to investigate neurotransmitter concentration through the skull, indicating the identification of melatonin, serotonin, and epinephrine through a cat skull at various dosage down to 100 μ M [17,28]. Principal components analysis (PCA), an unstructured multivariate detection method which decreases

the amplitude in large data sets, was used to indicate that SESORS could clearly investigate the individual neurotransmitters through the skull. The PCA analysis also denotes that the spectra acquired were distinctive for each neurotransmitter, resulting in a clear detachment of the numerous concentration spectra into clusters. These outcomes pave the way for possibilities in vivo neurotransmitter diagnosis through the skull. Integrating SERS with multivariate analysis permits for subtle dissimilarities in spectra due to variation between different molecules or disease and normal tissue states to be explained, which builds up the ability of SERS for disease diagnosis. [27,28]

DIABETES DETECTION

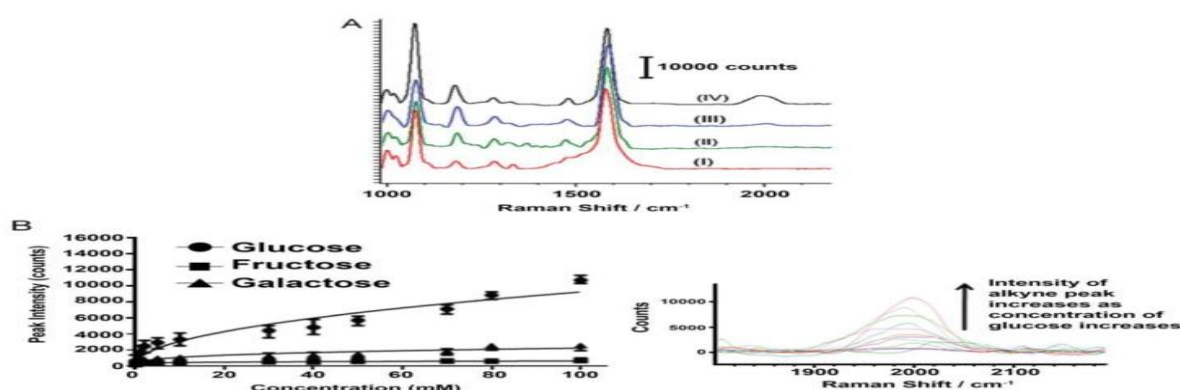


Fig 8: (A) Resemblance of SERS spectra of phenylboronic acid-operationalized substrate with (I) no analyte, and 10 millimole of (II) glucose, (III) fructose, and (IV) galactose; (B) SERS potency of alkyne peak at 1996 cm^{-1} with different dosage of glucose, fructose, and galactose. The Raman spectra is showing the alkyne summit with different dosage of glucose from Ref. [31]

In recent times, much progress has been made regarding the evolution of a biosensor that utilizes Raman spectroscopy to observe diabetes in real time. The most used glucose scanner need blood to be pinched from the finger and laid down on a strip, where a reaction occurs between the glucose oxidase (GOx) and blood glucose, and an electrode estimates the potency of the electric field produced. The newly obtainable CGM sensors exploit the same assumption with the tampering of the electrode being loaded subcutaneously [29]. In case of most current diabetes SERS-based sensors uses the catalytic venture of GOx in the oxidation of glucose to gluconic acid and hydrogen peroxide. A scientist enhanced a substrate protected with AgNPs and operationalized with GOx to act as a secondary glucose sensor chip when excited with a laser [31]. In the existence of more glucose, less SERS signal is prevailed because the GOx separates from the substrate [29]. The benefits of this probe include its high preciseness to glucose and that it is non-toxic. Non-invasive SERS identification of hydrogen peroxide, an outcome of glucose oxidation, has also been explored as a secondary indicator of glucose concentration. Utilizing a boronic acid probe, another scientist have identified a positive relationship between the SERS signal intensity acquired from hydrogen peroxide and the concentration of glucose in human serum and artificial urine. It has been denoted that the

glucose levels in saliva and body fluids are proportional to blood glucose levels [30]. Another scientist have created a perfect SERS substrate for the assembly of the hydrogen peroxide signal in bodily fluids beside blood utilizing AuNPs protected with silica, which allows for higher water solubility and also avert the reaction of gold with peroxide, and then operationalized the silica-protected AuNPs with GOx [31]. This sensor can correctly quantify glucose in vitro in the range from 25 μ M to 25 mM and is flexible to the introduction of glucose to saliva. A scientist has developed a more complicated substrate for direct glucose investigation that makes use of numerous of the methods [29,30]. The procedure demands coating AgNPs with an Au shell, then GOx, pursued by mercaptophenyl boronic acid to selectively arrest glucose and the obtain SERS signal. A linear scale curve was secured through in vitro estimations in the range from 2 mM to 6 mM, and the perfect quantification of glucose was obtained in blood samples. The sensor was decided to be stable for at least 30 days and supply selectivity for glucose when coupled in a matrix comprising of potentially interfering proteins.

It is also significant to note the recent innovations in in vitro SERS techniques for diabetes diagnosis. By performing PCA on the SERS spectra secured from forty urine samples, another scientist was able to discern between healthy and diabetes patients [30,31]. The differences in the intensity of definite peaks between the two groups correspond to familiar phenomena, such as the smaller amount of uric acid or the greater quantity of albumin present in the biofluids of diabetic patients. Again, a scientist used SERS to document how the construction of oxyhemoglobin changes with the evolution of diabetes [29]. The spectral differences of the blood samples permit the group to identify healthy patients versus diabetic patients.

From the direct diagnosis of glucose by boronic acid arrest label molecules, ligands, and sandwiching to the indirect sensing of glucose through its catalyzed oxidation, there are many chances to implement SERS in the clinical management of glucose levels. These methods are likely to progress the scope, longevity, and convenience of continuous glucose checking technology and enhance the lives of diabetic patients. [29]

CARDIOVASCULAR DISEASE DETECTION

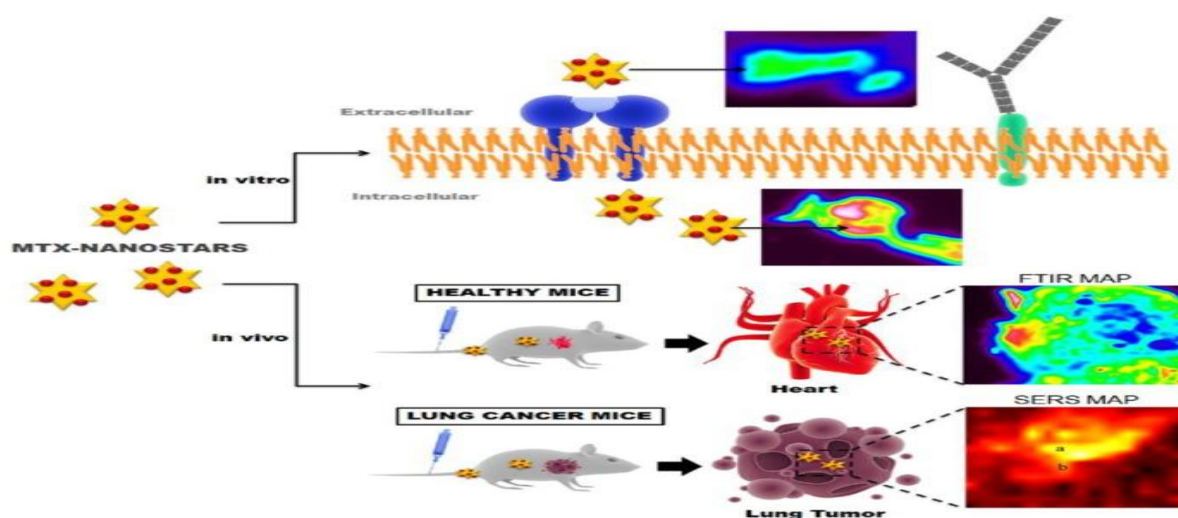


Fig 9: Explanation of the use of mitoxantrone (MTX) nanostars for in vivo and in vitro imaging, as well as non-definite remedy for patients with cardiovascular diseases from Ref. [32].

SERS is very constructive for the quantification and identification of biomarkers of cardiovascular diseases, such as uric acid. A scientist have used Fourier transform (FT)-Raman to identify uric acid with colloidal SERS substrates [18]. Three varieties of AgNPs were produced which caused reduction of citrate, hydroxylamine hydrochloride, and Ag seed-catalyzed ascorbic acid. The hydroxylamine hydrochloride reduced nanoparticles resulted in the considerable SERS achievement. One problem with this diagnosis technique, however, is the variation between the band positions in the SERS spectrum of uric acid on the hydroxylamine hydrochloride Ag nanoparticles compared to the normal Raman spectra, producing complications in identifying uric acid in biological fluids. The considerable changes observed are an outcome of uric acid molecules deprotonating/tautomerizing from communication with the silver surface and in the moist solution. C-reactive protein (CRP) is another indicator that has recently been observed as a potential marker of cardiovascular disease, using an Au nanoparticle-based SERS bioassay for diagnosis [32]. In this bioassay, AuNPs are operationalized with an antibody for CRP, along with horseradish peroxidase (HRP) and leucomalachite green (LMG). LMG is not a powerful Raman scattering molecule; however, when subjected to H_2O_2 through the conjugated HRP, it is lowered and converted to malachite green (MG), which displays a powerful SERS signal. The MG SERS signals altered in intensity in response to different concentrations of CRP, which suggests that the assay has positive selectivity and sensitivity for detecting CRP. N-terminal pro-brain natriuretic peptide (NT-proBNP) which is a delicate cardiac biomarker for heart failure and has been diagnosed with SERS using a sandwich assay. In vivo SERS diagnosis techniques have been evolved to detect and assisted with drug delivery in patients with cardiovascular diseases [19]. Healthy mice with lung cancer were provided gold nanostars conjugated with Mitoxantrone (MTX-nanostars), an anticancer drug, which has also been used effectively as a non-specific anti-inflammatory remedy for patients with chronic heart failure or cardiovascular diseases producing a SERS dynamic probe. Delivery of the drug was spotted in real-time using SERS diagnosis and Raman/FTIR imaging. In the healthy mice, in vivo SERS diagnosis found that the MTX-nanostars assembled in the heart. In vivo SERS spectra of the tumors in the mice with lung cancer also revealed featured SERS peaks linked with aromatic stretching of C-C (1300 cm^{-1}) of the MTX-nanostars. While originally evolved as a technique for detecting drug delivery to tumors, the MTX-nanostar probes have also been pertained to diagnosis of cardiovascular diseases due to the non-specific gathering of the MTX-nanostars in the hearts of healthy mice. Atherosclerosis is a cardiovascular disease which creates plaque to develop in arteries. SERS has been utilized for both treatment and early detection of this condition [18,19]. Vascular cell adhesion molecule-1 (VCAM-1) is believed to portray the development of atherosclerosis, so the identification of VCAM-1 could be used to diagnose atherosclerosis even before clinical symptoms emerge. SERS dynamic AuNPs were operationalised with 4-mercaptobenzoic acid (4-MBA) and the 4-MBA spectrum was detected in human coronary artery endothelial cells after VCAM-1 was overexpressed, showing the use of this SERS nanoprobe for biomarker perception. [32]

VIRAL DISEASES DETECTION

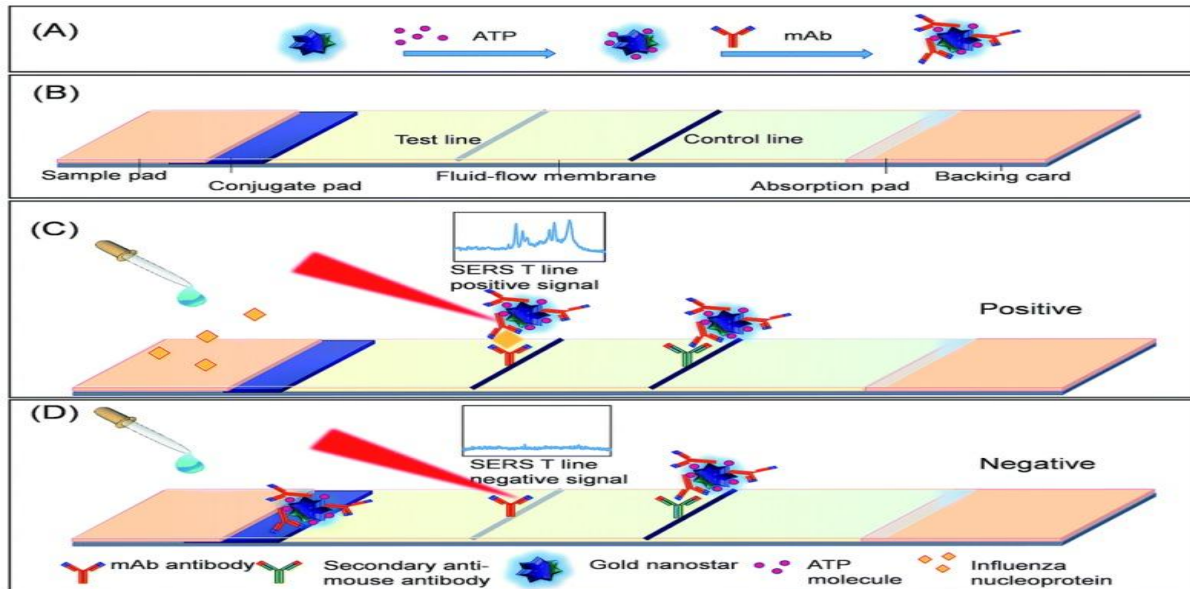


Fig 10: SERS-LFIA analysis system: (A) AuNS is operationalized with 4-ATP and integrated to monoclonal antibodies to produce the SERS tags; (B) LFA analysis strip, and the feedback of the SERS-LFIA arrangement to (C) influenza A nucleoprotein and (D) a negative dominance from Ref. [22]

SERS has been applied to detect viruses including Hepatitis, influenza, HIV and those answerable for tropical diseases. A constructed SERS substrate, comprising of multilayer hybrid Au/Ag nanorods presented in an array, was fused with rhodamine 6G (R6G) and utilized as a molecular probe. This nanoprobe displayed an analytical magnified factor of $\sim 10^7$ and diagnosis of Influenza A virus strains (H2N2, H3N2, and H1N1) with an LOD of 10^6 plaque-forming units (PFU) mL^{-1} . A scientist [15] have made a SERS-based lateral flow immunoassay (LFIA), applying influenza A nucleoprotein as a selected molecule. Au nanospheres (AuNS) were prepared following the technique of Frens [22], followed by the expansion of multibranched gold nanostars, which were linked to 4-ATP (AuNS-ATP). Antibodies were subsequently integrated to the AuNS-ATP. Samples comprising the target analyte were administered to LFIA strips, where nucleoprotein-NP complexes drifted across the strips to be arrested by the antibody integrated to the NPs and SERS spectra were taken at the position of gathering on the strip. The LOD was denoted to be 6.7 ng mL^{-1} . Samples of allantoic fluid counting influenza A (H1N1) were also analyzed and supported the validity of the technique to test for the virus in a matrix. It has also been studied that commercially obtainable diagnostic kits for rapid identification of influenza A can show upgraded diagnosis with SERS-based modifications [25]. Gold nanoparticles in a commercial diagnostic LFA kit were exchanged with AuNPs linked with Raman reporter malachite green isothiocyanate (MGITC). The technique measured influenza A with high sensitivity, with an LOD of $1.9 \times 10^4 \text{ PFU mL}^{-1}$, which was one order of magnitude less than the commercial LFA kit. This in

vitro diagnostic technique is a potential mixture to the resolution and sensitivity issues experienced when utilizing commercial kits. [24]

Hepatitis is a major source of liver diseases, which includes liver cancer, ensuing the need for a diagnosis technique. The maximum spotting techniques often need complicated sample synthesis and are expensive, while surface enhanced techniques for the diagnosis of hepatitis B virus (HBV) can be conveyed with little to no sample synthesis and in a cost-efficient fashion. There is a requirement for the sensitive spotting of human immunodeficiency virus (HIV), which would permit the earlier detection of HIV infection. This initial diagnosis would permit the start of anti-retroviral treatment earlier to severe immune consequences, and diagnosis via SERS has developed an attractive technique due to the sensitivity afforded by the method. A SERS-based HIV-1 DNA LFA biosensor, comprised of gold nanoparticles labelled with Raman reporter MGITC, was integrated to thiolated ssDNA complementary to HIV-1 [15,22]. HIV-1 DNA was quantitatively examined with SERS, yielding an LOD of 0.24 pg mL^{-1} , and this outcome was compared with a commercial kit, which had a much greater LOD (80 pg mL^{-1}). This SERS-based viewpoint also indicated three orders of magnitude lower LOD than colour intensity and fluorescent diagnosis techniques. Mosquito-borne viruses such as West Nile (WNV), dengue (DENV) and Zika (ZIKV) have been a subject of research in recent years, due to outburst of the diseases. Due to the existence of similar earlier symptoms, similar means of infection, and phenomenon in the same geographic regions, it is hard to differentiate between the two viral mosquito-imparted tropical viruses ZIKV and DENV. A scientist [33] evolved a SERS-based immunoassay with multifold spotting of the two viral diseases by using SERS-encrypted Au nanostars integrated to antibodies specific to each virus, and with separate Raman reporter molecules in an immunoassay to differentiate between nonstructural protein 1 (NS1) biomarkers. LODs for the NS1 protein of ZIKV and DENV were 0.72 ng mL^{-1} and 7.67 ng mL^{-1} , sequentially. A scientist [33] have also detailed a bio-conjugated anti-flavivirus 4G2 antibody-Au nanoparticle-based SERS probe to observe DENV and WNV. This investigation displayed that the AuNPs can be used as a probe to observe DENV-2 and WNV with an LOD of 10 PFU mL^{-1} . This LOD is equivalent to those typically seen with quantitative-PCR, but without complicated sample synthesis and processing. The technique was tested against a negative control of mosquito-borne chikungunya virus, with no Raman signal acquired from the negative control. Due to spectral variation between each virus, the technique was also able to differentiate between the two viruses. [34]

Laboratory identification of dengue is significant for delivering proper supportive therapy to dengue patients with febrile illness, which is hard to detect clinically. Here, we establish surface enhanced Raman scattering (SERS) based detection of dengue virus in clinical blood specimen gathered from total of 102 subjects. All the specimens were well featured by conventional NS1 antigen and IgM antibody ELISA kits. The silver nanorods array constructed by glancing angle deposition method were employed as SERS substrates. A small amount of patient blood serum ($5 \text{ }\mu\text{L}$) was taken for examination and the review was prepared within a minute. [34] SERS spectra of pure NS1 protein as well as spiked in serum was also recorded separately. Principal component analysis (PCA) was applied as the statistical apparatus to distinguish dengue positive, dengue negative, and healthy subjects based on their

respective SERS spectra. This technique supplies a rapid, sensitive, and field deployable diagnosis of dengue at the initial stage (within 5 days of the arrival of symptoms). [25]

CANCER DETECTION

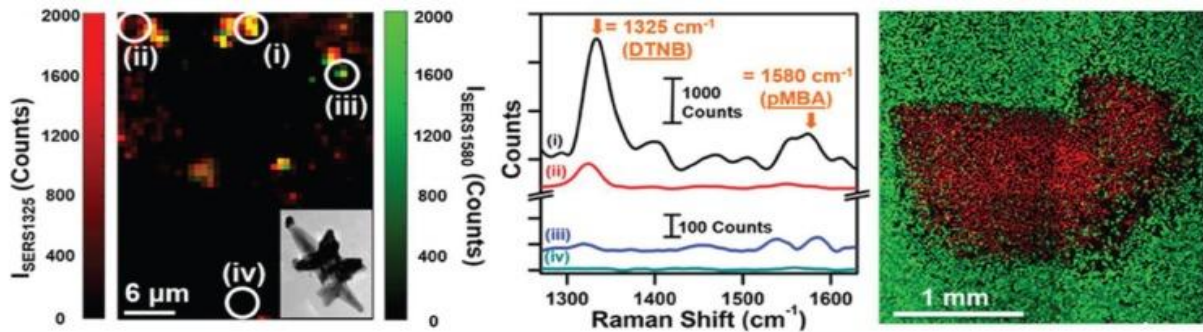


Fig 11: (Left) Raman potency representation of ISERS1325 and ISERS1580 acquired from diverse visualization of MDA-MB-231 TNBC cells using several branched gold nanoantennae (inset) exhibiting signals from (i) both probes, (ii) DTNB only, (iii) 4-MBA only, and (iv) no signal; (middle) the Raman spectra acquired from (i) both probes, (ii) DTNB alone, (iii) 4-MBA alone, and (iv) no probes or intracellular lipids; and (right) living cells (green) and deceased cells (red) noticed via confocal fluorescence visualization after photothermal cell demise from Ref. [21]

SERS is an enchanting target for accomplishment in cancer diagnosis due to high sensitivity, little-to-no sample clarifying, and rapid detection times. The most lethal cancers in men include liver, colorectal, lung, prostate and stomach cancers; for women, the most lethal cancers are lung, colorectal, cervical, stomach and breast cancers. There has been important research interest in the epitome of SERS methods into the observation of cancer cells in serum, the observing of drug delivery to tumours, and the estimation of excision margins during the surgical removal of tumours. [16]

BREAST CANCER: A SERS-based technique termed Raman-encrypted molecular imaging (REMI) has been evolved to grant for the sensitive, rapid and specific resolution of cancer cell-positive surgical margins during tumour removal policies. REMI depends on the topical functions of SERS active nanoparticles operationalized with antibodies that are specific to a panel of cell-surface biomarkers, which are acquired from mastectomy and excised lumpectomy tissues. The antibodies utilized to target the biomarkers include membrane estrogen receptor (mER), epidermal growth factor receptor (EGFR), human epidermal growth factor receptor 2 (HER2), and CD44 [20]. EFGR is down-regulated in the cells of triple-negative breast cancer (TNBC), a highly fatal and aggressive models of breast cancer, and thus can be utilized as a biomarker for the observation of TNBC cells. A scientist have displayed a multiplexed diagnosis of both EFGR and programmed death ligand 1 (PGL1), which is also down regulated in multiple sub-lines of breast cancer cells [21]. To establish the multiplexed detection, two apparent Raman reporters 4-mercaptopbenzoic acid (4-MBA) and 5,5-dithio-bis-(2-nitrobenzoic acid) (DTNB) were engaged, with each reporter integrated to a

multibranched nanoantenna substrate and antibodies particular to the biomarker of interest. SERS diagnosis of the cancerous cells was combined with photothermal treatment to stimulate cell death within the laser spot. [20,21]

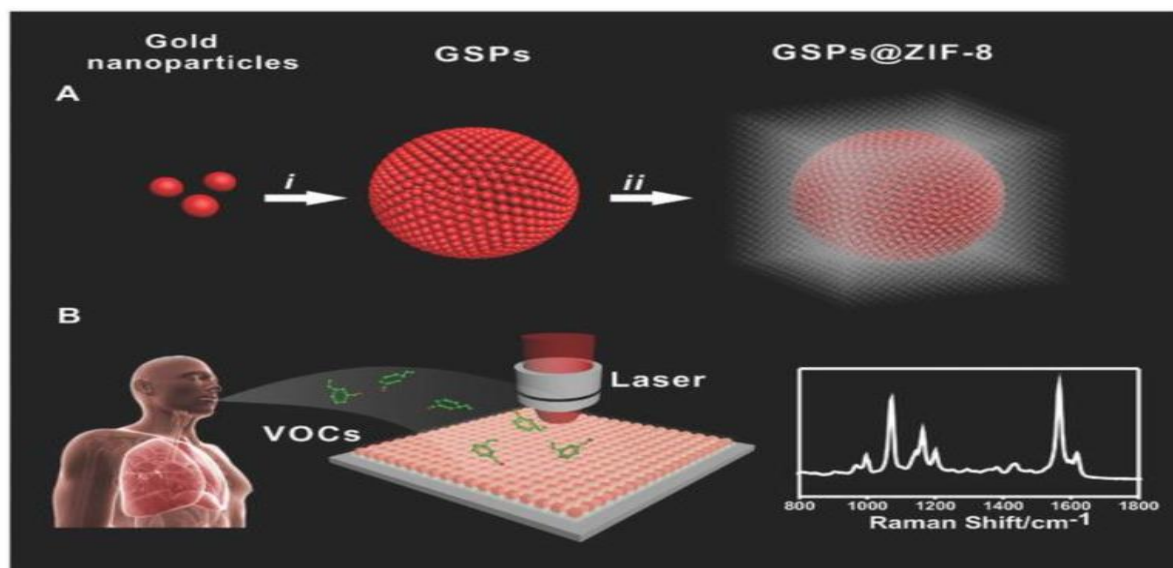


Fig 12: A) Illustrative sketch of preparing GSP@ZIF-8 core-shell design: (i) AuNPs fabricated into GSPs, (ii) ZIF-8 shell varnished on GSP surface; (B) volatile organic compound (VOC) diagnosis via SERS spectroscopy from Ref. [36]

LUNG CANCER: Tumors are familiar to exist in decreased pH condition than normal tissue, which make them principal victim for pH-based sensors. pH-sensitive SERS sensors have been evolved and executed as a technique to quantify the intercellular acidification of lung cancer cells focused on the modification of the initial substrate, 4-nitrothiophenol accumulated onto a gold nanorod, under hypoxic environment to 4-aminothiophenol (4-ATP), which is controlled by the pH of the cell. Gas analysis of volatile organic compounds is a rising policy for the non-invasive investigation of aldehydes that are feature of lung cancer. [36]. A scientist used a microemulsion assembly technique, wherein monodisperse (5.8 ± 0.6 nm mean diameter) Au nanoparticles were gathered into Au superparticles (GSP, 170 ± 30 nm diameter) with single-crystal-like attributes. These GSP cores were then operationalized with 4-ATP and protected with a ~ 150 nm porous ZIF-8 metal-organic-structure (MOF) to provide porosity and maintain electromagnetic fields around the substrate. By specifying the small molecular weight aldehydes via pore size exclusion, integrated with Schiff base reactions between aldehydes and amines, the sensor was able to observe gaseous aldehydes at the 10-ppb level [35]. The application of silver dendritic nanocrystals operationalized with 4-ATP using Schiff base reactions without the MOF layer has also been analysed and resulted in a similar ppb investigation of aldehydes.

ADDITIONAL CANCERS: Microfluidic lab-on-chip (LOC) appliances, SERS, and statistical examination have been merged to distinguish between three different leukemia cell lines, cultured from patients with severe lymphocytic leukemia and severe myelocytic leukemia (THP-1 and MONO-MAC-6 cells) [16]. SERS spectra were secured for the three cell lines, succeeded by PCA on the acquired spectra, resulting in well divided groupings

corresponding to each cell type. A support vector machine (SVM) model, a well guided regression and classification examination technique, was produced to arrange the cell types, resulting in mean specificities, sensitivities, and accuracies of >99% for all three cell types, based on >2000 spectra aggregate. SERS and PCR methods were united for the tandem multiplexed diagnosis of single-base abnormalities located in melanoma lesions to distinguish between three varieties of point mutations in tumor DNA [37]. This SERS-based genotyping was also adapted to mutant and wild type DNA acquired from three melanoma cell lines, with the outcomes established by droplet digital PCR (ddPCR). The SERS-genotyping was then applied to circulating tumor DNA (ctDNA) acquired from melanoma patients, with sensitivity and specificity similar to ddPCR methods but with simple PCR policies [16].

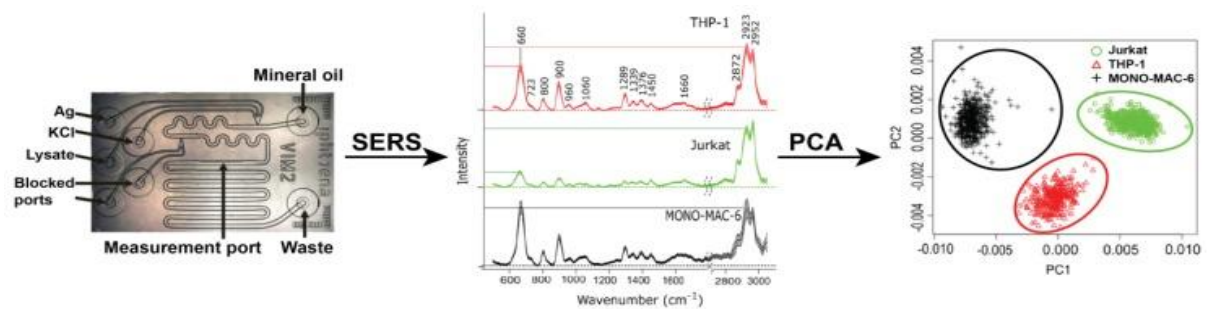


Fig 13: A quartz chip microfluidic apparatus was constructed for SERS at 785 nm excitation. At three distinct inlets of the microfluidic apparatus, AgNPs, KCl, and the cell lysate are introduced and flow to merge (left). Mineral oil is introduced to the initial mixture to act as a dissociation medium for the emergence of droplets. SERS measurements (middle) of the droplets are obtained at a midpoint in the apparatus. PCA of SERS spectra distinctly shows isolation and recognition of the three leukemia cell lysates utilizing variations in band potency between the spectra (right) from Ref. [37]

CONCLUSION AND FUTURE PROSPECT

SERS grants for the constant, extremely sensitive diagnosis and calibration of different biomarkers and end products of ailments states, which marks it as an excellent choice in the identification and treatment of many disorders that are global public health apprehension. With progress in instrumentation and as techniques for higher selectivity towards reducing LODs are evolved, SERS will remain an indispensable technique, and pursue to display great potential, for in vivo and in vitro disease diagnosis. The capability to establish substrates from gold, which is non-virulent during in vivo application, is a supplementary welfare supplied by SERS. SERS is likely to assist the scope, longevity and convenience of constant glucose checking technology, as well as assistance in earlier diagnosis of neurological, viral, cardiovascular diseases, and cancer. As LODs are reduced by the development of more highly strengthen substrates, more enlightened capture policies, and multimodal imaging techniques, it is superficial that SERS will resume to remain at the forefront of highly sensitive analysis in both in vivo and in vitro utilization.

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SERS is an enchanting target for accomplishment in cancer diagnosis due to high sensitivity, little-to-no sample clarifying, and rapid detection times. The most lethal cancers in men include liver, colorectal, lung, prostate and stomach cancers; for women, the most lethal cancers are lung, ovarian, breast and pancreatic cancers. There has been a lot of research on SERS methods into the detection of various biomarkers.

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